

Lagrangian Heuristic for the Facility Location Problem

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Abstract: In the two-stage capacitated facility location problem a single product is produced at some plants in order to satisfy customer demands. The product is transported from these plants to some depots and then to the customers. The capacities of the plants and depots are limited. The aim is to select cost minimizing locations from a set of potential plants and depots. This cost includes fixed cost associated with opening plants and depots, and variable cost associated with both transportation stages. In this work a Lagrangian relaxation is analyzed and a Lagrangian heuristic producing feasible solutions is presented. The results of a computational study are reported.

1. INTRODUCTION

The two-stage capacitated facility location problem can be defined as follows: a single product is produced at plants and then transported to depots, both having limited capacities. From the depots the product is transported to customers to satisfy their demands. The use of the plants/depots incurs a fixed cost, while transportation from the plants to the customers through the depots results in a variable cost. We need to identify what plants and depots to use, as well as the product flows from the plants to the depots and then to the customers such that the demands are met at a minimal cost.

The literature on facility location problem and its applications is quite extensive. We refer here only for a few recent review papers [8, 14, 24, and 26]. In what follows we focus more on approximate techniques to solve facility location problems. For exact approaches see, e.g., [1, 15] and the references therein.

Approximate approaches can be roughly divided into two large groups: metaheuristics and Lagrangian based techniques [30]. Metaheuristic approaches to the problem like tabu search, GRASP, are discussed in [9, 28]. A cross entropy-based metaheuristic algorithm for the capacitated facility location problem was studied in [5]. An algorithm for large instances is presented in [2], they used a heuristic procedure that produces a feasible integer solution and used a Lagrangian relaxation to obtain a lower bound on the optimal value.

A Lagrangian based heuristic for solving the capacitated plant location problem with side constraints was presented in [27]. Approaches and relaxations proposed in the literature for the capacitated facility location problem are compared in [7]. Fenchel cutting planes methodology was applied in [25] to capacitated facility location problems and compared the results obtained by a Lagrangian relaxation.

A linear programming based heuristic is considered in [12] for a two-stage capacitated problem with single source constraints. A hybrid algorithm was presented in [19], integrating the approximation approach, neural network and a simulated annealing to solve a capacitated fuzzy two-stage location-allocation problem. In [31] a greedy construction heuristic was proposed and combined with a Variable Neighborhood Descent and a Variable Neighborhood Search for the multi-stage facility location problem with staircase costs and splitting of commodities. In [16] the asymmetry inherent to the problem in plants and depots is taking into account to enforce the formulation. Two formulations for the problem are presented in [10], the linear relaxations are compared with the binary relaxation of the model.

Several Lagrangian relaxation approaches have been proposed for the two stage facility location problem. For the uncapacitated case in [6] they studied the effectiveness of the formulation for the two level simple plant location problem incorporating polyhedral cuts and of an approach combining a Lagrangian relaxation method and simulated annealing algorithm. In [20] an algorithm was proposed based on Lagrangian heuristics for a 0-1 mixed integer model of a two level location problem with three types of facility to be located. In [22] a mixed integer formulation and several Lagrangian relaxations to determine lower bounds for the two stage uncapacitated facility location problem are presented.

The Lagrangian relaxation for the capacitated case was studied and numerically tested in [3]. Alternative model formulations of the capacitated problem were studied in [4] obtaining lower bounds by Lagrangian relaxations of the flow-balancing constraints. They developed also heuristic procedures to obtain feasible solutions. Several Lagrangian relaxations for two different formulations of the two-stage problem are computationally compared in [21]. A Lagrangian relaxation-based branch and bound algorithm for the two-echelon, single source, capacitated problem was

proposed in [29]. A Lagrangian heuristic is proposed in [13] using relaxation of the capacity constraints for the problem with a fixed number of plants. Feasible solutions are constructed from those of the Lagrangian subproblems by applying simple reassignment procedures.

In many techniques Lagrangian relaxation is used in twofold: the optimal value of the Lagrangian (dual) problem is used as a dual bound, while the Lagrangian solution is used as a starting or reference point to produce a feasible solution and a corresponding primal bound. Frequently a relaxation is considered as *good* if it produces a tight dual bound. Meanwhile, the quality of the feasible Lagrangian based solution has also to be taken into account in evaluating the relaxation. There are often different ways in which a given problem can be relaxed in a Lagrangian fashion. It is unlikely to highlight a single relaxation producing high quality bounds of both types, primal and dual. Moreover, if the quality of the dual bound is basically defined by the constraints relaxed, the quality of the primal bound depends also on the algorithm used to restore the feasibility of the Lagrangian solution.

In this paper we consider a simple decomposable relaxation and an algorithm to restore the feasibility of the corresponding Lagrangian solution. This relaxation produces a poor dual bound and never been considered before as a promising relaxation. Meanwhile, the relaxation results in a very tight feasible solution typically within 0.5-1.0% of the relative suboptimality.

The rest of the paper is organized as follows. The next section presents a mathematical formulation for the two-stage facility location problem and the Lagrangian bound. A heuristic procedure to get feasible solutions is presented in section 3. Computational results are reported in section 4, while section 5 concludes.

2. PROBLEM FORMULATION AND LAGRANGIAN BOUND

To formally describe the problem, let $I = 1, \dots, n$ be the index set of potential plants, $J = 1, \dots, m$ the index set of potential depots and $K = 1, \dots, k$ the index set of clients. Then, the problem can be formulated as the following mixed integer linear program:

$$w = \min \sum_{i \in I} f_i y_i + \sum_{j \in J} g_j z_j + \sum_{i \in I, j \in J} c_{ij} x_{ij} + \sum_{j \in J, k \in K} d_{jk} s_{jk} \quad (1)$$

$$\sum_{j \in J} s_{jk} \geq q_k, k \in K, \quad (2)$$

$$\sum_{i \in I} x_{ij} \geq \sum_{k \in K} s_{jk}, j \in J, \quad (3)$$

$$\sum_{j \in J} x_{ij} \leq b_i y_i, i \in I, \quad (4)$$

$$\sum_{i \in I} x_{ij} \leq p_j z_j, j \in J, \quad (5)$$

$$x_{ij}, s_{jk} \geq 0, y_i, z_j \in \{0, 1\}; i \in I, j \in J, k \in K \quad (6)$$

Here f_i and g_j are the fixed costs associated with the installation of plant i and depot j ; c_{ij} and d_{jk} are the costs of transportation from plant i to depot j and from depot j to client k , respectively; q_k is the demand of client k ; while b_i and p_j are the capacities of the corresponding plant and depot. The variables in this formulation are $y_i = 1$ if plant i is installed and $y_i = 0$ otherwise, $z_j = 1$ if depot j is installed and $z_j = 0$ otherwise, x_{ij}, s_{jk} are the transportation flows between the corresponding units.

Constraint (2) is the demand constraint (for each customer, at least the demand must be met); (3) is the relaxed flow conservation constraint (the product transported from the depot must at least be transported to it from the plants); constraints (4) and (5) represent capacity limits for plants and depots and assure the flow only from plants and depots opened. Note, that by minimizing the objective (1), constraints (2) and (3) are fulfilled as equalities for an optimal solution of (1)-(6).

Lagrangian bounds are widely used as a core of many numerical techniques, e.g. in branch and bound schemes for integer and combinatorial problems, see [11] and the references therein. Most Lagrangian relaxation approaches for the capacitated facility location problem are based either on dualizing the demand constraints or the depot capacity constraints [12]. In this paper we consider the relaxation of the demand and the plant capacity constraints (2) and (4), respectively. For fixed multipliers $u, v \geq 0$ the corresponding Lagrangian problem is stated as follows:

$$\begin{aligned} L(u, v) = \min & \sum_{i \in I} f_i y_i + \sum_{j \in J} g_j z_j + \sum_{i \in I, j \in J} c_{ij} x_{ij} + \sum_{j \in J, k \in K} d_{jk} s_{jk} \\ & + \sum_{k \in K} u_k (q_k - \sum_{j \in J} s_{jk}) + \sum_{i \in I} v_i (\sum_{j \in J} x_{ij} - b_i y_i) \\ & \sum_{i \in I} x_{ij} \geq \sum_{k \in K} s_{jk}, j \in J, \\ & \sum_{i \in I} x_{ij} \leq p_j z_j, j \in J, \end{aligned}$$

$$x_{ij}, s_{jk} \geq 0, y_i, z_j \in \{0, 1\},$$

providing the lower bound $L(u, v) \leq w$. Rearranging the terms in the objective function yields:

$$L(u, v) = \sum_{k \in K} q_k u_k + \min \left\{ \sum_{i \in I} (f_i - b_i v_i) y_i + \sum_{j \in J} [g_j z_j + \sum_{i \in I} (c_{ij} + v_i) x_{ij} + \sum_{k \in K} (d_{jk} - u_k) s_{jk}] \right\}$$

Thus the Lagrangian problem decomposes into I independent subproblems in y_i , $L_i(v_i) = \min_{y_i \in \{0, 1\}} (f_i - v_i b_i) y_i$, that are solved analyzing the sign of the coefficients $(f_i - v_i b_i)$, and J independent subproblems each having only one binary variable z_j :

$$\begin{aligned} L_j(u, v) = \min & \left\{ g_j z_j + \sum_{i \in I} (c_{ij} + v_i) x_{ij} + \sum_{k \in K} (d_{jk} - u_k) s_{jk} \right\} \\ & \sum_{i \in I} x_{ij} \geq \sum_{k \in K} s_{jk}, \end{aligned}$$

$$\sum_{i \in I} x_{ij} \leq p_j z_j, \\ x_{ij}, s_{jk} \geq 0, z_j \in \{0, 1\}.$$

The later problem is analyzed by inspection, fixing z_j to 0 or 1 and then solving the remaining (network flow) problem in continuous variables (x, s) .

Thus we may conclude that the computational cost to solve the Lagrangian problem corresponding to the relaxation of the demand and the plant capacity constraints is very low, in fact no integer problem is involved. In [18] another equivalent formulation for the problem (1)-(6) was considered and 10 different Lagrangian relaxations were compared numerically for both formulations. We refer the reader to [18] for details of this numerical study. The problem of finding the best, i.e. bound maximizing Lagrange multipliers, is called the Lagrangian dual. To solve the Lagrangian dual problem one can apply a subgradient technique. Another popular approach is the constraint generation scheme (Benders method) transforming the dual problem into a large-scale linear programming problem. The advantage of using Benders technique is that it generates two-sided estimations for the dual bound in each iteration thus producing a near-optimal dual bound with guaranteed quality. Meanwhile, the computational cost of this scheme is typically high. In contrast to the Benders method, the subgradient technique does not provide the value of the bounds with the prescribed accuracy. That is, terminating iterations of the subgradient method using some stopping criteria we can expect only approximate values of the bound. We do not consider here these two well known approaches in details, referring the reader to [17, 23] for the constraint generation (Benders) technique, and to [23, 32] for the subgradient scheme. Note also that in [18] the bounds were calculated by both techniques and the results obtained by the subgradient technique were very close to those obtained by Benders technique.

3. GETTING FEASIBLE SOLUTIONS

To get a feasible solution from the Lagrangian one we use a simple algorithm to recover feasibility. In fact, this approach can be applied to any nonfeasible solution.

Algorithm

Let $\bar{x}_{ij}, \bar{s}_{jk}$ be a non-feasible solution

Do $y_i = 0, \forall i, I_1 = \emptyset, I_0 = I; z_j = 0, \forall j, J_1 = \emptyset, J_0 = J$.

Step 0: Do $y_i \leftarrow (\sum_{j \in J} \bar{x}_{ij}) / b_i, z_j \leftarrow (\sum_{k \in K} \bar{s}_{jk}) / p_j$.

Step 1: $i^* = \arg \max \{y_i | i \in I_0\}$

Step 2: $y_{i^*} \leftarrow 1, I_1 \leftarrow I_1 \cup \{i^*\}, I_0 \leftarrow I_0 - \{i^*\}$.

Step 3: If $\sum_{i \in I_1} b_i \geq \sum_{k \in K} q_k$ go to step 4 and do $y_i = 0, \forall i \in I_0$, otherwise, return to step 1.

Step 4: $j^* = \arg \max \{z_j | j \in J_0\}$.

Step 5: $z_{j^*} \leftarrow 1, J_1 \leftarrow J_1 \cup \{j^*\}, J_0 \leftarrow J_0 - \{j^*\}$.

Step 6: If $\sum_{j \in J_1} p_j \geq \sum_{k \in K} q_k$ go to step 7 and do $z_j = 0, \forall j \in J_0$, otherwise, return to step 4.

Step 7: Fix y_i and z_j in the original problem and solve corresponding linear problem to obtain the flows.

In this algorithm we calculate for each plant a “saturation” indicator representing the relative usage of its capacity (step 0). Then the plant having the highest saturation is opened (step 1). If the capacity is sufficient to satisfy the total customers' demand, the rest of the plants are closed, otherwise the plant having the next highest indicator is opened too. (Steps 2 and 3). The depots are opened in a similar way (steps 3, 5 and 6). Fixing the binary variables obtained by this procedure, the flows are determined from the corresponding linear problem.

4. COMPUTATIONAL RESULTS

A numerical study for the two-stage capacitated facility location problem was conducted to compare the bounds. The following sets of instances were generated according to the values (I; J; K): A(3; 5; 9); B(5; 7; 30); C(7; 10; 50); D(10; 10; 100); E(10;16;30); F(30;30;30); G (30;36;120); H(30;30;100); I (50;50;200). Every set contains 20 problem instances. The data were random integers generated as follows:

$$c_{ij}, d_{jk} \in U[10, 20], q_k \in U[1, 10],$$

$$b_i \in \left[10 \frac{J+K}{I}\right] + U[0, 10], p_j \in \left[10 \frac{K}{J}\right] + U[0, 10]$$

Two different ways to generate the fixed costs were implemented. For the first ten instances in each class the fixed costs f_i, g_j were random integers generated independently on the number clients, plants and depots: $f_i, g_j \in U[100, 200]$. For the remaining ten instances the fixed costs f_i for plants were proportional to the number of depots and clients, while the fixed costs g_j for depots were proportional to the number of clients:

$$f_i \in \left[100 \frac{K+J}{I} \right] + U[0,100]; g_j \in \left[100 \frac{K}{J} \right] + U[0,100]$$

The dual bound corresponding to the Lagrangian relaxation was calculated by the subgradient technique. In each iteration of this method the feasible solution was obtained by the Algorithm. The best (over all iterations) feasible solution was stored. The current best feasible solution was used to update the step size. If after 5 consecutive iterations of the subgradient technique the dual bound was not improved, the half of the step size scaling parameter was used. The process stops if the step size scaling parameter is less than 0.0001, or if the maximum number (300) of iterations is reached.

The procedure was implemented in GAMS/CPLEX 11.2 using a Sun Fire V440 terminal, connected to 4 processors Ultra SPARC III with 1602 Hhz, 1 MB of CACHE, and 8 GB of memory. For all the instances we have calculated: z_{IP} - the value of the optimal objective of the two stage location problem; z_L - the value of the best Lagrangian bound; z_{BF} - the objective value corresponding to the best feasible solution. The relative quality of the Lagrangian bound and of the best feasible solution was measured by the corresponding

$$\varepsilon_L = 100\%(z_{IP} - z_L)/z_{IP}, \quad \varepsilon_{BF} = 100\%(z_{BF} - z_{IP})/z_{BF}.$$

TABLE I. INSTANCES A-C, FIRST TYPE

Size	ε_L (%)	ε_{BF} (%)	ε_{LF} (%)	ε_{FF} (%)
A1	12.30	1.81 (14)	5.07 (135)	12.51
A2	2.31	0.00 (31)	0.00 (89)	14.95
A3	7.74	0.57 (5)	4.21 (138)	4.21
A4	6.39	0.00 (9)	0.00 (128)	11.63
A5	11.03	0.00 (4)	3.79 (130)	3.79
B1	6.04	0.00 (27)	0.99 (124)	3.47
B2	4.03	0.00 (17)	0.00 (135)	2.35
B3	5.93	1.26 (16)	4.73 (132)	2.66
B4	4.97	0.00 (15)	4.91 (147)	4.18
B5	4.15	0.18 (28)	0.18 (102)	8.37
C1	4.89	0.00 (36)	1.78 (118)	3.39
C2	4.43	0.35 (9)	2.81 (116)	8.22
C3	2.88	0.03 (8)	2.28 (151)	7.59
C4	4.59	0.00 (28)	6.57 (116)	8.49
C5	3.03	0.00 (28)	2.43 (118)	6.91

TABLE II. INSTANCES D-F, FIRST TYPE

Size	ε_L (%)	ε_{BF} (%)	ε_{LF} (%)	ε_{FF} (%)
D1	2.89	0.00 (16)	4.26 (113)	5.30
D2	2.43	0.00 (24)	3.14 (129)	7.44
D3	3.75	0.00 (96)	5.55 (131)	9.71
D4	3.70	0.49 (1)	2.16 (130)	4.19
D5	2.07	0.00 (105)	1.75 (153)	3.79
E1	4.56	0.00 (59)	0.10 (225)	11.65
E2	5.03	0.53 (92)	0.75 (217)	7.79

E3	3.94	1.37 (49)	5.67 (231)	12.86
E4	3.47	0.52 (134)	2.05 (256)	9.14
E5	4.25	0.46 (191)	6.07 (256)	9.10
F1	2.21	0.75 (109)	1.44 (300)	13.92
F2	1.89	0.00 (53)	0.89 (235)	13.67
F3	1.88	0.037 (58)	1.17 (198)	15.48
F4	1.61	1.55 (264)	3.67 (300)	19.74
F5	1.91	0.55 (60)	1.66 (266)	13.71

TABLE III. INSTANCES G-I, FIRST TYPE

Size	ε_L (%)	ε_{BF} (%)	ε_{LF} (%)	ε_{FF} (%)
G1	1.10	0.37 (240)	1.91 (300)	8.11
G2	0.99	1.18 (88)	1.58 (300)	7.63
G3	1.13	0.21 (249)	1.24 (300)	8.69
G4	0.94	0.56 (100)	1.81 (300)	7.42
G5	0.67	0.23 (194)	0.87 (300)	9.92
H1	1.04	0.46 (284)	2.02 (300)	10.28
H2	1.29	0.47 (168)	1.04 (300)	5.84
H3	1.49	0.38 (191)	1.57 (300)	8.18
H4	1.39	0.77 (284)	2.57 (300)	7.16
H5	1.73	0.73 (147)	3.93 (300)	8.83
I1	0.96	0.52 (199)	1.27 (300)	5.98
I2	0.93	0.81 (130)	2.52 (300)	6.44
I3	0.78	1.09 (196)	1.49 (300)	7.01
I4	0.73	0.92 (202)	1.57 (300)	5.94
I5	0.64	1.04 (94)	1.90 (300)	6.44

Tables 1-3 present the results obtained for the first way to generate data, while Tables 4-6 give the results for the second. The results are shown for 5 different instances for each problem size. The first two columns present the proximity indicators for the corresponding dual bound (ε_L) and for the best (over all iterations) feasible solution (ε_{BF}) obtained by the Algorithm. The last two columns give the proximity indicator for the feasible solution corresponding to the last (ε_{LF}) and the first (ε_{FF}) iteration of the subgradient technique. The number in the parenthesis indicates the number of the iteration corresponding to the bound value.

TABLE IV. INSTANCES A-C, SECOND TYPE

Size	ε_L (%)	ε_{BF} (%)	ε_{LF} (%)	ε_{FF} (%)
A1	13.64	0.51 (9)	4.72 (214)	10.31
A2	4.79	0.00 (29)	0.00 (166)	14.48
A3	5.33	0.43 (2)	13.46 (158)	18.41
A4	6.89	0.00 (28)	10.17 (188)	11.98
A5	19.09	0.00 (1)	0.00 (157)	0.00
B1	13.18	0.00 (137)	0.79 (212)	2.09
B2	9.09	0.00 (58)	1.62 (250)	1.44
B3	8.29	1.57 (22)	9.79 (250)	8.63
B4	5.49	0.00 (61)	6.01 (243)	6.01
B5	9.32	0.11 (21)	0.11 (252)	14.79
C1	6.68	0.00 (62)	8.28 (223)	9.95
C2	10.83	0.25 (181)	2.11 (270)	5.06

C3	6.84	0.56 (76)	2.28 (258)	4.64
C4	8.55	0.00 (27)	4.80 (237)	5.28
C5	6.47	0.00 (47)	1.49 (250)	4.34

TABLE V. INSTANCES D-F, SECOND TYPE

Size	ε_L (%)	ε_{BF} (%)	ε_{LF} (%)	ε_{FF} (%)
D1	6.32	0.20 (62)	2.12 (284)	6.79
D2	6.08	0.40 (237)	4.34 (256)	4.48
D3	8.48	0.00 (67)	1.62 (277)	5.64
D4	7.29	1.91 (18)	8.18 (250)	8.45
D5	5.57	0.19 (153)	3.58 (276)	2.19
E1	4.04	2.54 (44)	9.06 (227)	13.75
E2	9.56	0.79 (71)	0.79 (205)	4.93
E3	3.71	5.76 (51)	6.83 (293)	14.73
E4	6.25	0.00 (108)	1.69 (233)	7.35
E5	7.40	0.32 (56)	8.03 (277)	8.79
F1	3.15	0.67 (133)	1.31 (257)	13.41
F2	2.94	0.00 (60)	1.14 (261)	15.65
F3	2.28	0.95 (56)	2.93 (250)	14.24
F4	2.90	0.93 (51)	2.06 (261)	13.58
F5	2.96	0.64 (32)	5.42 (277)	20.07

TABLE VI. INSTANCES G-I, SECOND TYPE

Size	ε_L (%)	ε_{BF} (%)	ε_{LF} (%)	ε_{FF} (%)
G1	3.33	0.61 (194)	1.12 (300)	6.94
G2	1.98	0.82 (128)	1.56 (300)	8.22
G3	1.27	0.46 (120)	2.60 (300)	8.29
G4	0.87	0.16 (260)	1.11 (300)	7.15
G5	1.51	0.70 (87)	1.87 (300)	9.14
H1	1.64	1.16 (206)	2.52 (300)	8.33
H2	2.23	0.58 (214)	2.88 (300)	8.28
H3	1.54	0.23 (240)	2.25 (300)	7.26
H4	1.47	0.58 (127)	3.49 (300)	5.37
H5	1.14	0.43 (78)	2.64 (300)	8.81
I1	1.09	0.79 (211)	1.09 (300)	6.61
I2	1.95	0.81 (207)	2.27 (300)	5.62
I3	1.59	1.41 (221)	1.81 (300)	5.28
I4	1.29	1.01 (168)	2.89 (300)	6.59
I5	1.48	0.94 (146)	1.78 (300)	5.67

As we can see from the Tables for both ways to generate the data the approach provides very tight feasible solutions, typically within 0.5-1.0% of the relative proximity.

The figure 1 shows the proximity of the Lagrangian bounds in the course of the subgradient iterations for an instance of size A.

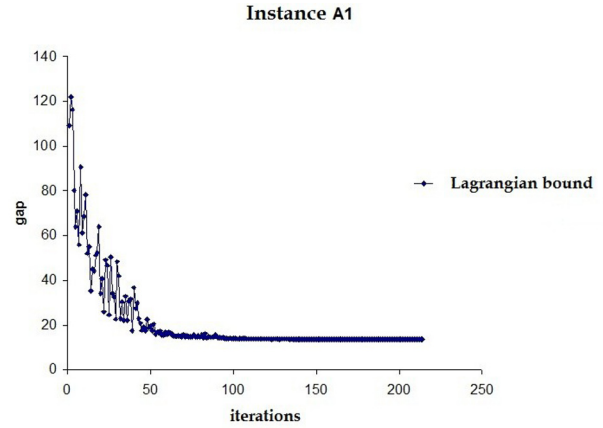


Figure 1. Proximity of the Lagrangian bounds

So we may expect that the population of the Lagrangian solutions generated by the subgradient technique in the course of solving the Lagrangian dual is “sufficient” for the Algorithm to generate high quality feasible solutions. The quality of the dual bound is poor, improving for larger instances.

The feasible solution derived from the solution of the Lagrangian dual and corresponding to the last iteration of the subgradient technique not necessarily is the best feasible solution, as we can see from the figure 2 which shows the proximity of the feasible solution in the course of the subgradient iterations for an instance of size A. Moreover, typically $\varepsilon_{LF} > \varepsilon_{BF}$ and the best feasible solution is obtained on the early iterations of the subgradient method. Thus we may conclude that it is important to generate feasible solutions in all iterations of the subgradient technique.

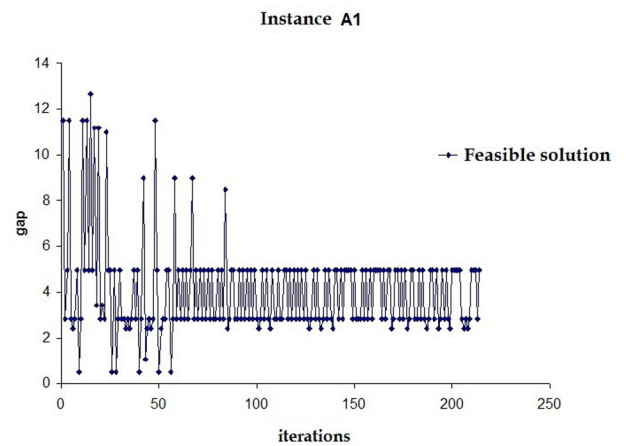


Figure 2. Proximity of the feasible solution

5. CONCLUSIONS

In many Lagrangian heuristics the optimal value of the Lagrangian (dual) problem is used as a dual (lower) bound for the optimal objective, while the Lagrangian solution is used as a starting or reference point for a heuristic to produce a feasible solution (upper primal bound). This way the proximity of the feasible solution to the optimal one is estimated. It is assumed implicitly that the best dual bound and the corresponding Lagrangian solution produces the best primal bound by a heuristic approach. Thus, the feasible solution is constructed using heuristics (e.g., greedy approaches) based on the Lagrangian solution obtained for the optimal Lagrange multipliers corresponding to the best dual bound.

One of the core ideas of the metaheuristic approaches is recognizing the role of “near optimal” solutions. That is, instead of using for the next iteration the locally optimal solution obtained for the current iteration, near optimal solutions are also taken into account to define the next iteration. Metaheuristic approaches using this idea were successfully applied for many hard combinatorial and integer problems.

We propose to combine the use of near optimal solutions with the Lagrangian approach. We demonstrate that the best primal solution typically is generated by the Lagrangian solution which is far from optimal. That is, if we apply an iterative approach (e.g., subgradient technique) to solve the Lagrangian dual, the Lagrangian solutions corresponding to current (non optimal) multipliers result in better feasible solutions.

Lagrangian bound was presented for the two stage capacitated facility location problem. Two indicators were considered: the quality of the dual bound and the proximity of the Lagrangian based feasible solution. It turned out that relaxing the demand and the capacity plant constraints provides a rather poor dual bound, but the Lagrangian based feasible solutions are good, typically within 0.5-1.0% of the relative suboptimality. Relaxing the demand and the capacity plant constraints result in a decomposable Lagrangian problem with all subproblems analyzed by inspection. Thus this low cost relaxation seems to be promising to form the core of the Lagrangian based heuristics.

Solving the dual problem by the subgradient technique we compared two approaches to generate a feasible Lagrangian based solution. One is to get a feasible one by the solution of the dual problem, i.e. at the last iteration of the subgradient technique. Another approach is to generate feasible solutions in all iterations of the subgradient method and then choose the tightest. It turned out that the best (over all iterations) feasible solution never was obtained at the last iteration. That is, simply solving the Lagrangian dual and getting a corresponding Lagrangian based feasible solution is not sufficient to produce a tight feasible solution. On the contrary, the population of the Lagrangian solutions generated by the subgradient technique in the course of solving the Lagrangian dual is “sufficient” to generate high quality feasible solutions.

An interesting direction for the future research is improving the heuristic used to derive the feasible solutions. Some complements in this direction are in course.

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